
Monetary policy topics

Adrien Auclert

Goethe Heterogeneous-Agent Macro Workshop, 2026

This session

- ❖ So far: HANK with Jacobians by hand, introduction to the SSJ toolkit
- ❖ **Now:** more advanced topics on monetary policy, using the toolkit!
 1. Cyclical income risk
 2. Maturity structure
 3. Nominal assets
 4. The stock market

Cyclical income risk

Cyclical income risk

- ❖ Recall canonical model: household takes $n_{it} = N_t$ as given and solves

$$\max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(n_{it}) \right)$$

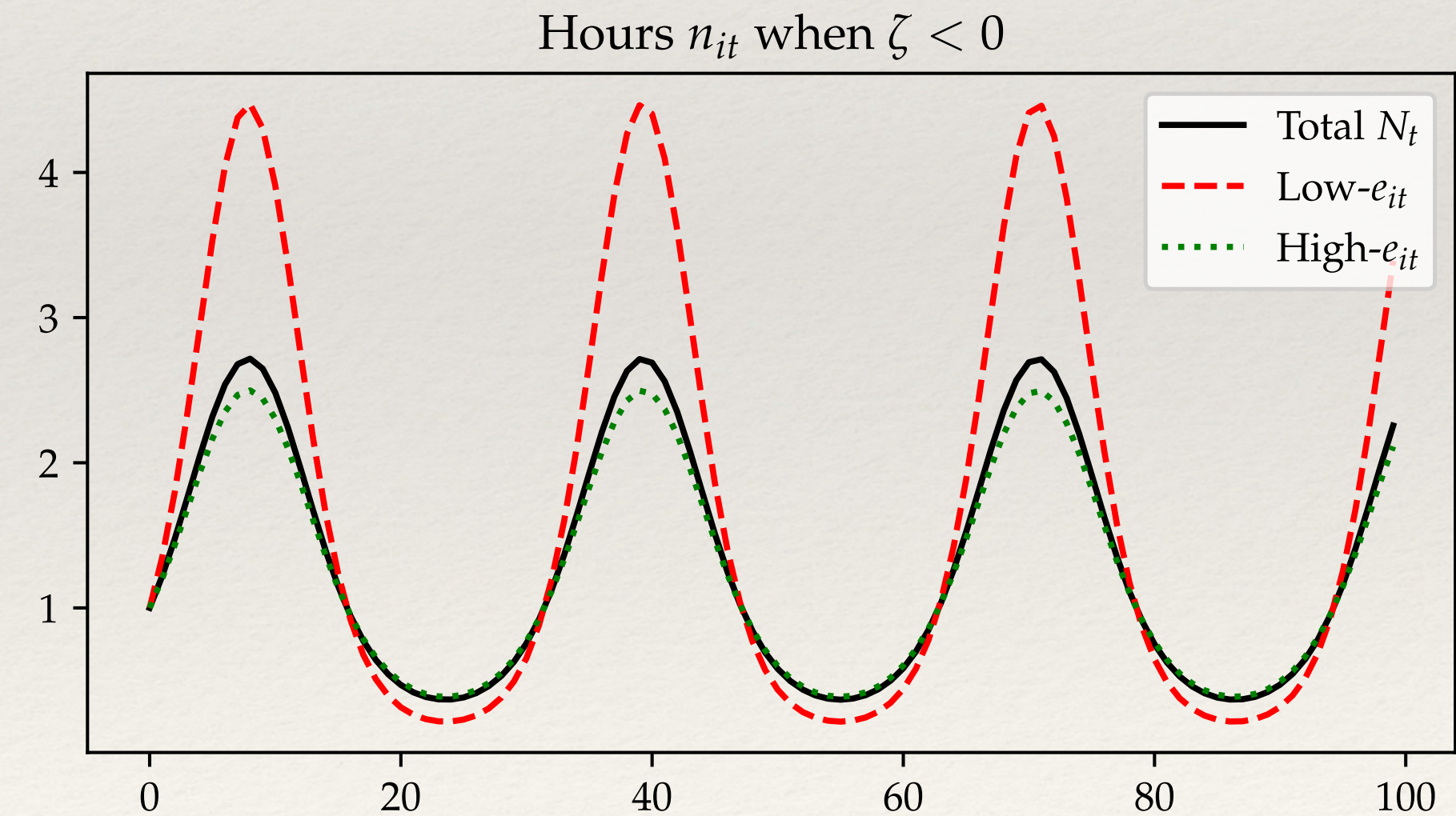
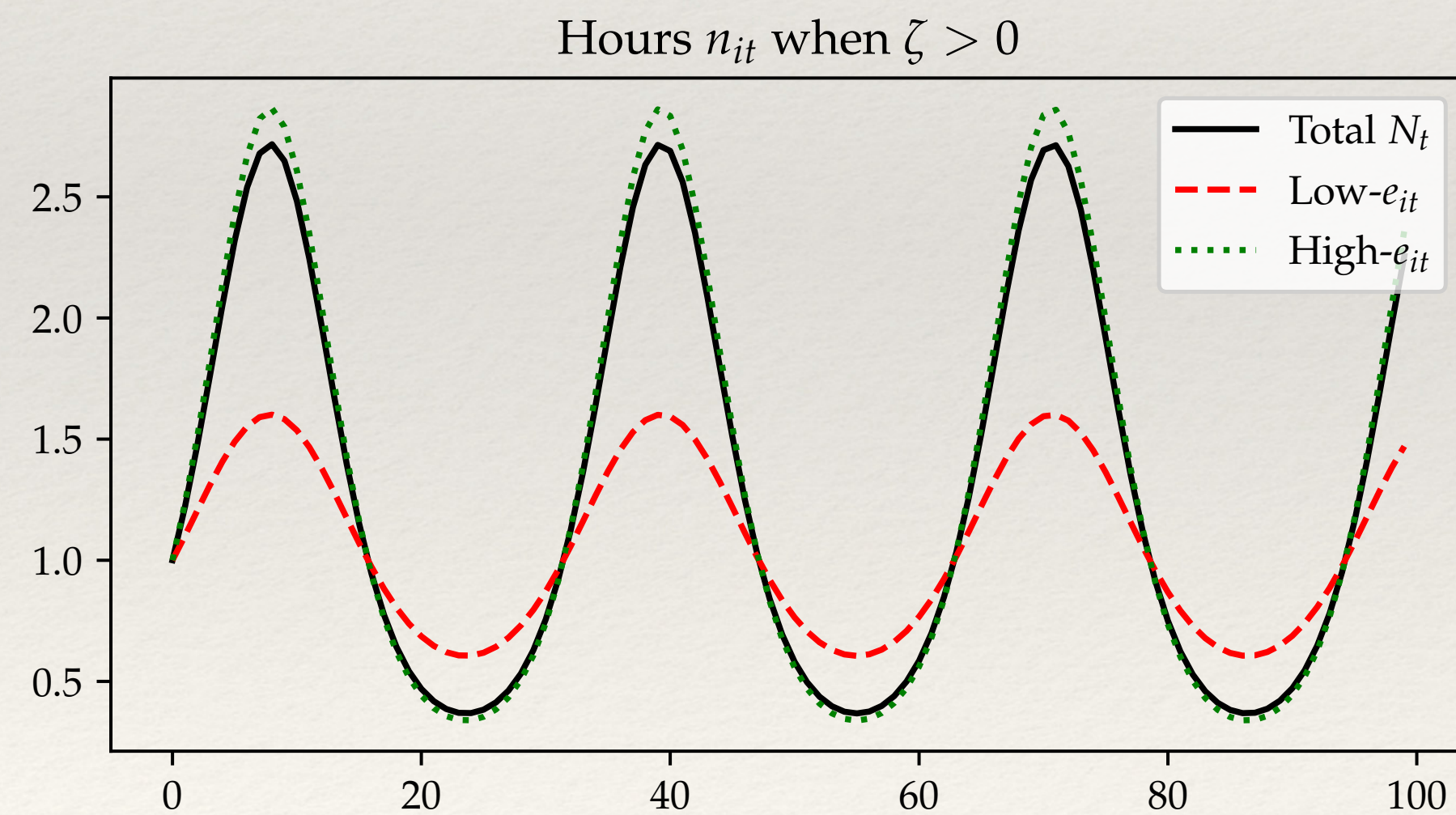
$$c_{it} + a_{it} \leq (1 + r_t^p) a_{it-1} + (1 - \tau_t) \frac{W_t}{P_t} n_{it} e_{it}$$
$$a_{it} \geq 0$$

- ❖ This is restrictive: hours n_{it} don't move uniformly across people over the cycle!
- ❖ Relaxing this assumption will make a difference, but also uncover a big “puzzle”

Introducing unequal rationing

- ❖ Simple way to relax is to assume that labor is rationed as a function of e_{it} , per
(assuming normalization $N_{ss} = 1$)

$$n_{it} = N_t \frac{(e_{it})^{\zeta \log N_t}}{\mathbb{E} \left[e_i^{1+\zeta \log N_t} \right]} \equiv N_t \Gamma(e_{it}, N_t)$$



- ❖ $\zeta > 0$: procyclical inequality+income risk, $\zeta < 0$ countercyclical, $\zeta = 0$ acyclical

Introducing unequal rationing

- ❖ Simple way to relax is to assume that labor is rationed as a function of e_{it} , per
(assuming normalization $N_{ss} = 1$)

$$n_{it} = N_t \frac{(e_{it})^{\zeta \log N_t}}{\mathbb{E} \left[e_i^{1+\zeta \log N_t} \right]} \equiv N_t \Gamma(e_{it}, N_t)$$

- ❖ $\zeta > 0$: procyclical inequality+income risk, $\zeta < 0$ countercyclical, $\zeta = 0$ acyclical

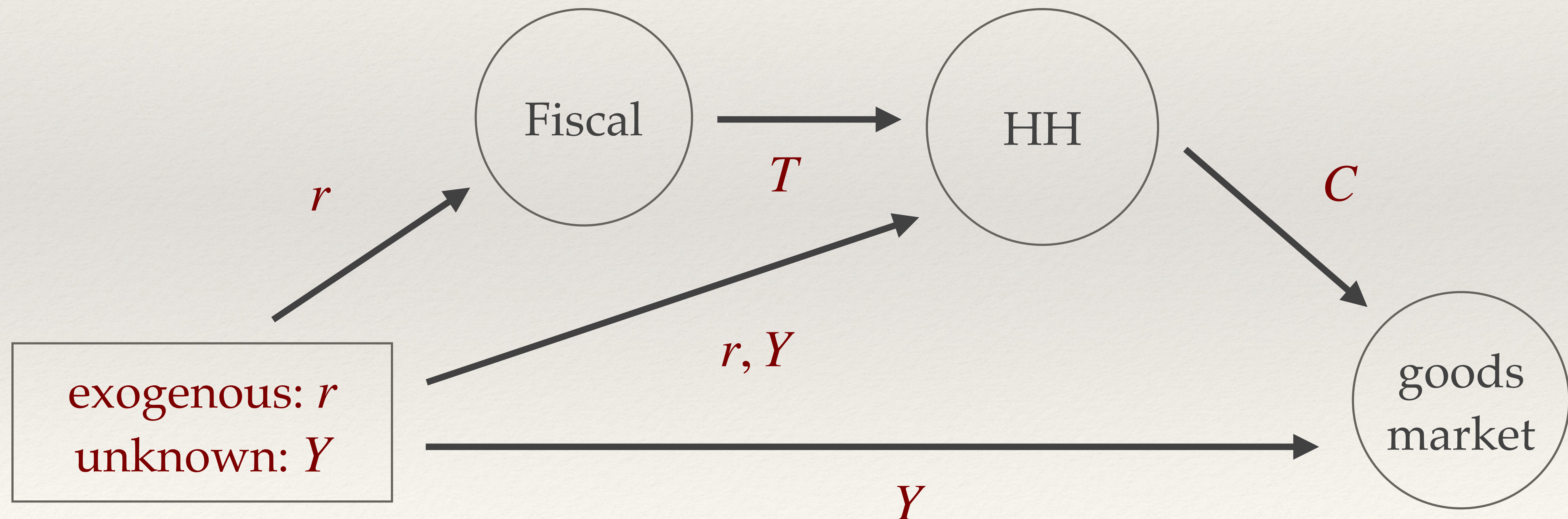
- ❖ In particular the distribution of pretax income $y_{it} = \frac{W_t}{P_t} n_{it} e_{it}$ moves with the cycle

$$\text{sd}(\log y_{it}) = (1 + \zeta \log N_t) \text{sd}(\log e_i)$$

- ❖ Matters because: 1) current shocks redistribute between different MPCs, and
2) future shocks change perceived income risk

How does this change the model?

- ❖ Now post-tax income z_{it} for individual i is $z_{it} = (Y_t - T_t)\Gamma(e_{it}, Y_t)$
- ❖ This results in a consumption function $\mathcal{C}_t\left(\{r_s, Y_s, T_s\}\right)$. DAG:



Practical implementation with hetinput:

- ❖ Simple modification of the “hetinput” to the household block:

```
def income_cyclical(Y,T, e_grid, e_pdf, zeta):
    y = (Y-T) * e_grid ** (1 + zeta * np.log(Y)) / np.vdot(e_grid ** (1 + zeta * np.log(Y)), e_pdf)
    return y
```

```
household_cyc = hh.add_hetinputs([income_cyclical])
```

```
print('Household block inputs:' , household_cyc.inputs)
print('Household block outputs:' , household_cyc.outputs)
```

✓ 0.0s

Python

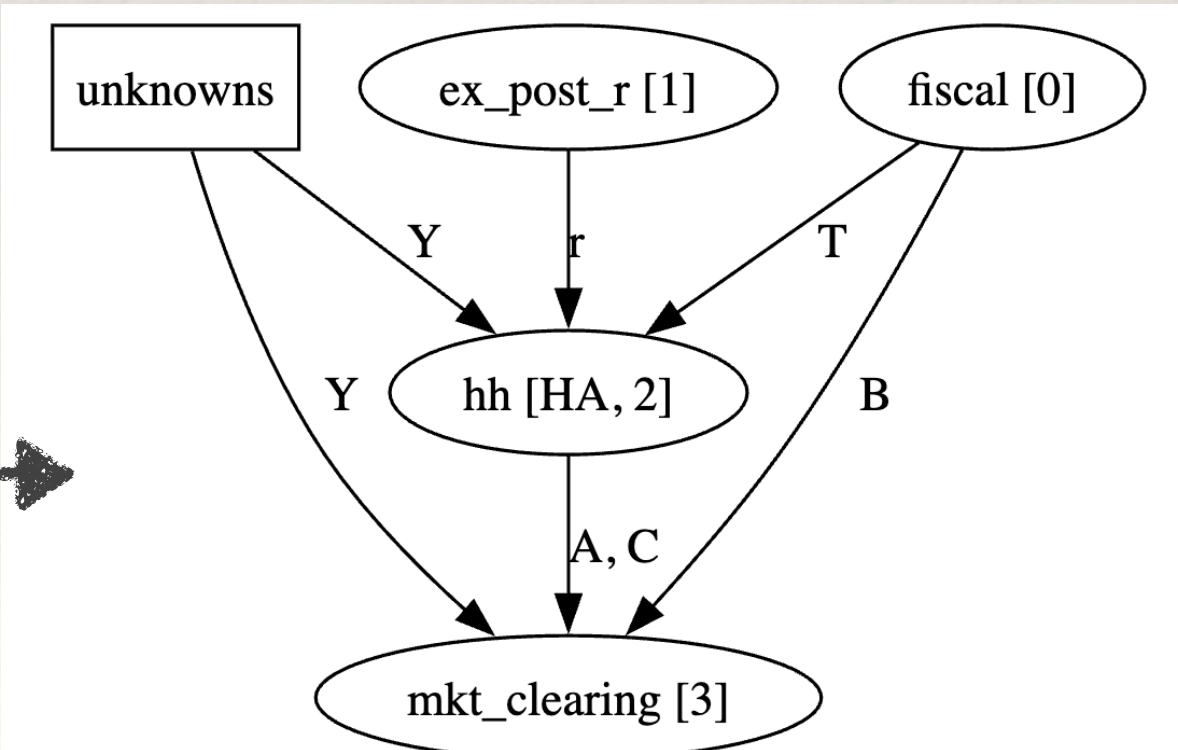
Household block inputs: ['a_grid', 'r', 'beta', 'eis', 'Pi', 'Y', 'T', 'e_grid', 'e_pdf', 'zeta']

Household block outputs: ['A', 'C']

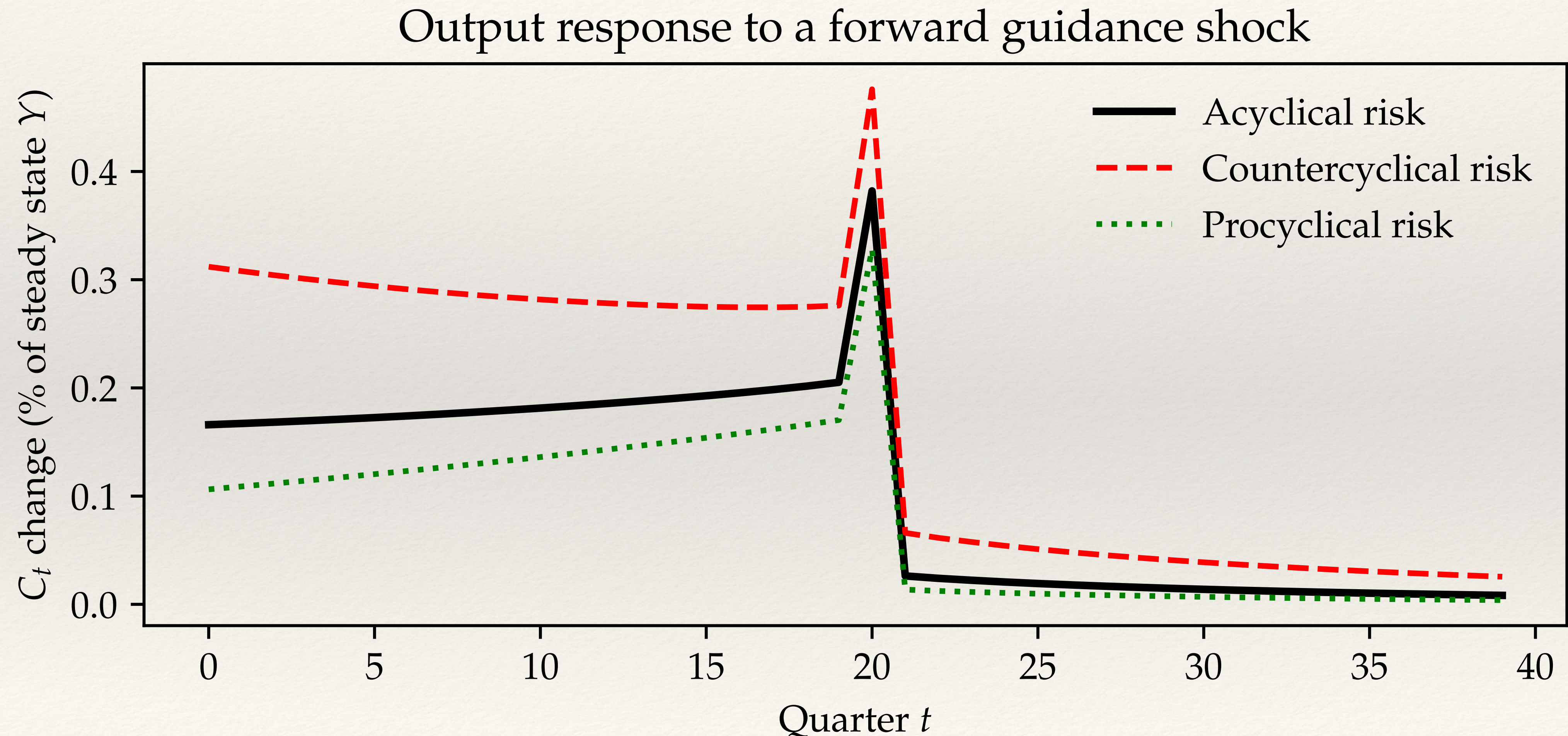
- ❖ Can then combine this and other blocks into model:

```
model_cyc = sj.create_model([household_cyc, ex_post_r, fiscal, mkt_clearing], \
                             name="HA Model with cyclical income risk")
```

```
sj.drawdag(model_cyc, unknowns=['Y'])
```



Result: countercyclical risk amplifies FG puzzle!



What is happening?

- ❖ Go back to zero liquidity limit $B_{ss} \rightarrow 0$. Now, can show:

$$C_t = \delta \cdot C_{t+1} - \frac{1}{\sigma} \cdot \text{const} \cdot (r_t - \bar{r})$$

- ❖ There is dynamic discounting $\delta < 1$ iff procyclical income risk $\zeta > 0$
- ❖ Dynamic amplification $\delta > 1$ iff countercyclical income risk $\zeta < 0$
- ❖ Countercyclical income risk usually thought to be more plausible:
 - ❖ Theoretically: eg, from unemployment risk (Ravn-Sterk, Challe, Kekre...)
 - ❖ Empirically: eg, work by Guvenen&co and Storesletten&co
- ❖ Bilbiie's "catch-22": countercyclical risk plausible at micro level, not macro!

Indirect ways to make income risk cyclical

- ❖ In richer models, income of agents typically involves multiple components,

$$y_{it} = \frac{W_t}{P_t} n_{it} e_{it} - \underbrace{T_{it}}_{\text{Taxes}} + \underbrace{Tr_{it}}_{\text{Transfers}}$$

- ❖ These also matter for cyclicity of income risk
- ❖ For example, suppose we set taxes to keep $(1 + r_t)B_t$ constant as in our baseline, but that transfers are (nontraded) dividends from firms with sticky prices
 - ❖ Both T_{it} and Tr_{it} fall when $r_t \downarrow$ (why?)
- ❖ Suppose that T_{it} are paid by the high e_{it} , but everyone owns equal share of firms
- ❖ Then income risk is procyclical! This is McKay, Nakamura, Steinsson

Maturity structure

Maturity structure

- ❖ So far: agents trade short term assets. What if longer maturities / duration?
- ❖ For tractability, assume “Calvo bonds”
 - ❖ buy one bond today for q_t , coupon payments $1, \delta, \delta^2, \dots$

- ❖ New household problem:
$$\max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(N_t) \right)$$
$$c_{it} + q_t \lambda_{it} \leq (1 + \delta q_t) \lambda_{it-1} + Z_t e_{it}$$
$$\lambda_{it} \geq 0$$

Total number of coupons owned coming into t



- ❖ Asset pricing condition:
$$1 + r_t = \frac{1 + \delta q_{t+1}}{q_t}$$

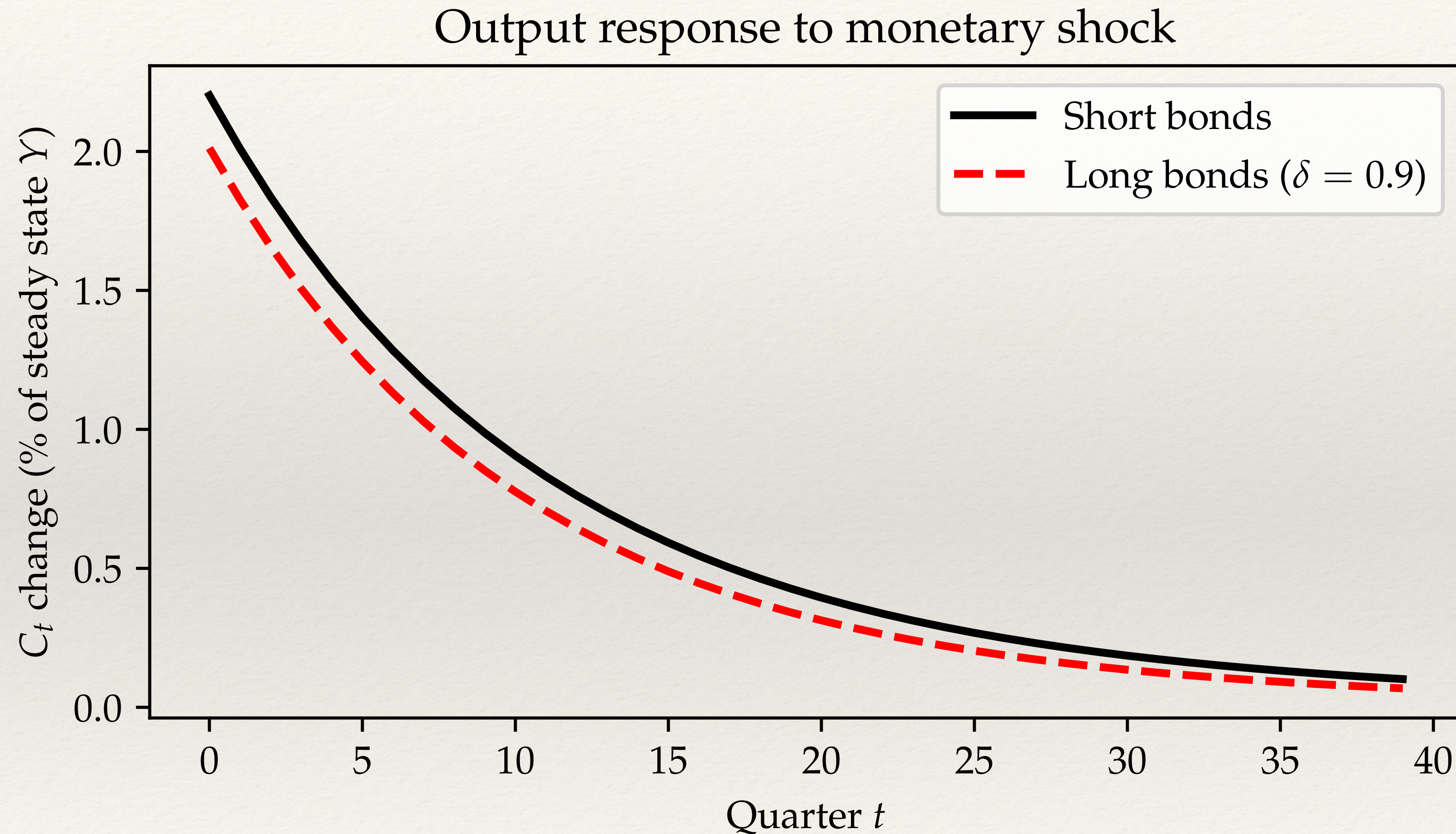
Rewriting the problem

- ❖ Redefine $a_{it} = q_t \lambda_{it}$. Using asset pricing condition, we have

$$\begin{aligned} \max_{c_{it}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(N_t) \right) & r_{t+1}^p &= r_t, t \geq 0 \\ c_{it} + a_{it} & \leq (1 + r_t^p) a_{it-1} + Z_t e_{it} & 1 + r_0^p &= \frac{1 + \delta q_0}{q_{ss}} \\ a_{it} & \geq 0 \end{aligned}$$

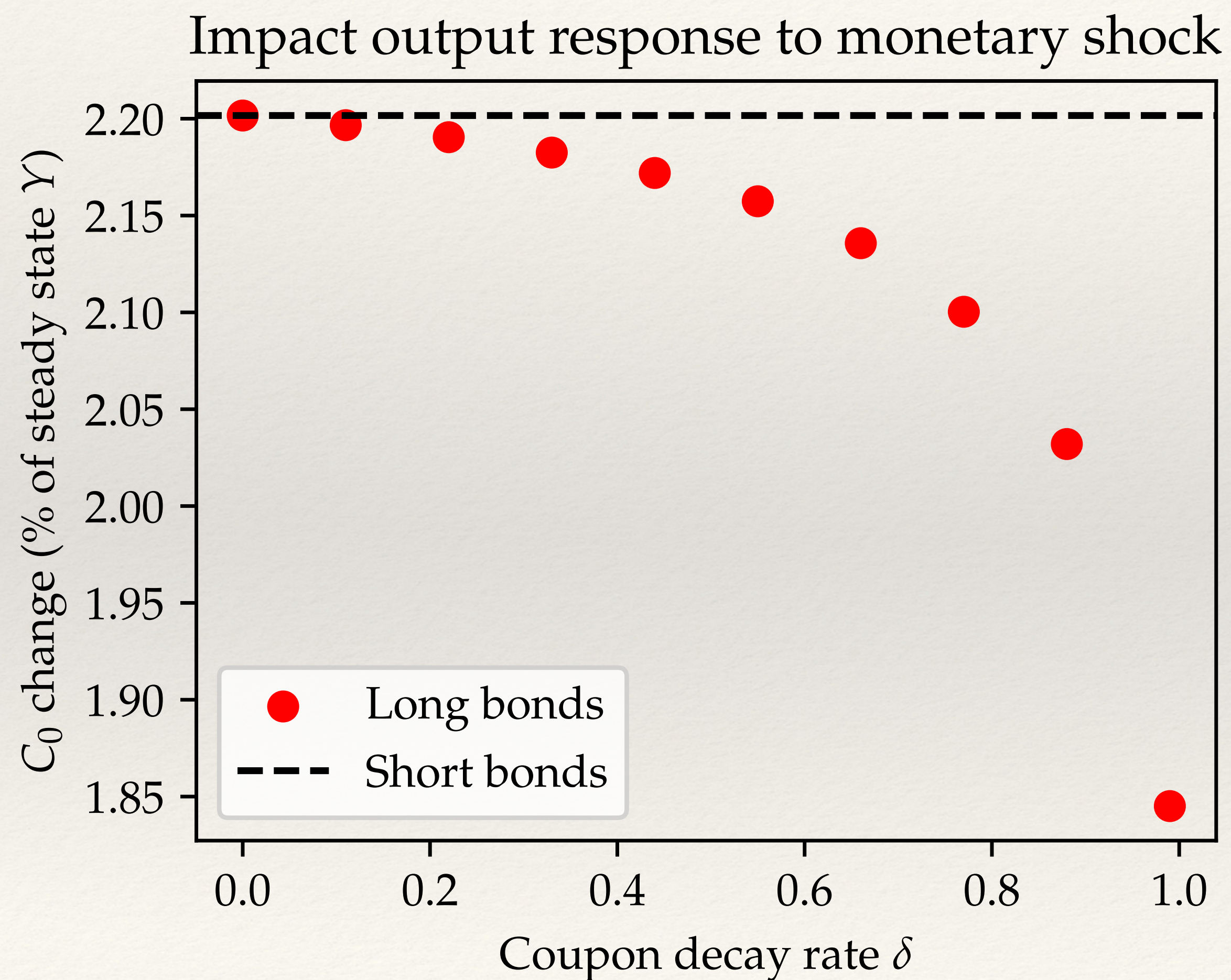
- ❖ Lower ex-ante $r_t \rightarrow$ higher bond price $q_0 \rightarrow$ higher r_0^p (capital gain)
- ❖ Same for government budget constraint: $B_t = (1 + r_t^p) B_{t-1} + G_t - T_t$
 - ❖ here $B_t \equiv q_t \lambda_t$ is market value of gov debt. Assume **fiscal rule** $\lambda_t = \lambda_{ss}$, $G_t = G_{ss}$

Effect of long maturities on monetary impulse



❖ Longer maturities reduce the output effect of a monetary shock ! Why?

Effect of maturity structure explained



- ❖ $\delta \uparrow$: households make cap. gains from monetary expansion, gov makes loss
- ❖ Redistribution from taxpayers to bond holders. Latter have lower MPCs, so aggregate effect on spending falls!
- ❖ Expect monetary policy to be more powerful in countries with short durations (eg ARMAs)
- ❖ Actually true in data! (Calza et al)

Nominal assets

Nominal assets

- ❖ So far, all assets are real. But in practice, many assets are nominal
 - ❖ Mortgages, bonds, etc
 - ❖ Creates very large exposures to inflation risk via nominal positions (see e.g. Doepke and Schneider)

- ❖ Consider 1-period bonds:
$$\max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(N_t) \right)$$
$$P_t c_{it} + A_{it} \leq (1 + i_t) A_{it-1} + P_t Z_t e_{it}$$

$$A_{it} \geq 0$$

Can relax to have $P_t \underline{a}$

- ❖ Fisher equation:

$$1 + r_t = \frac{(1 + i_t) P_t}{P_{t+1}}$$

In practice, borrowing constraints interact with inflation in complex ways! (“Tilt effect”)

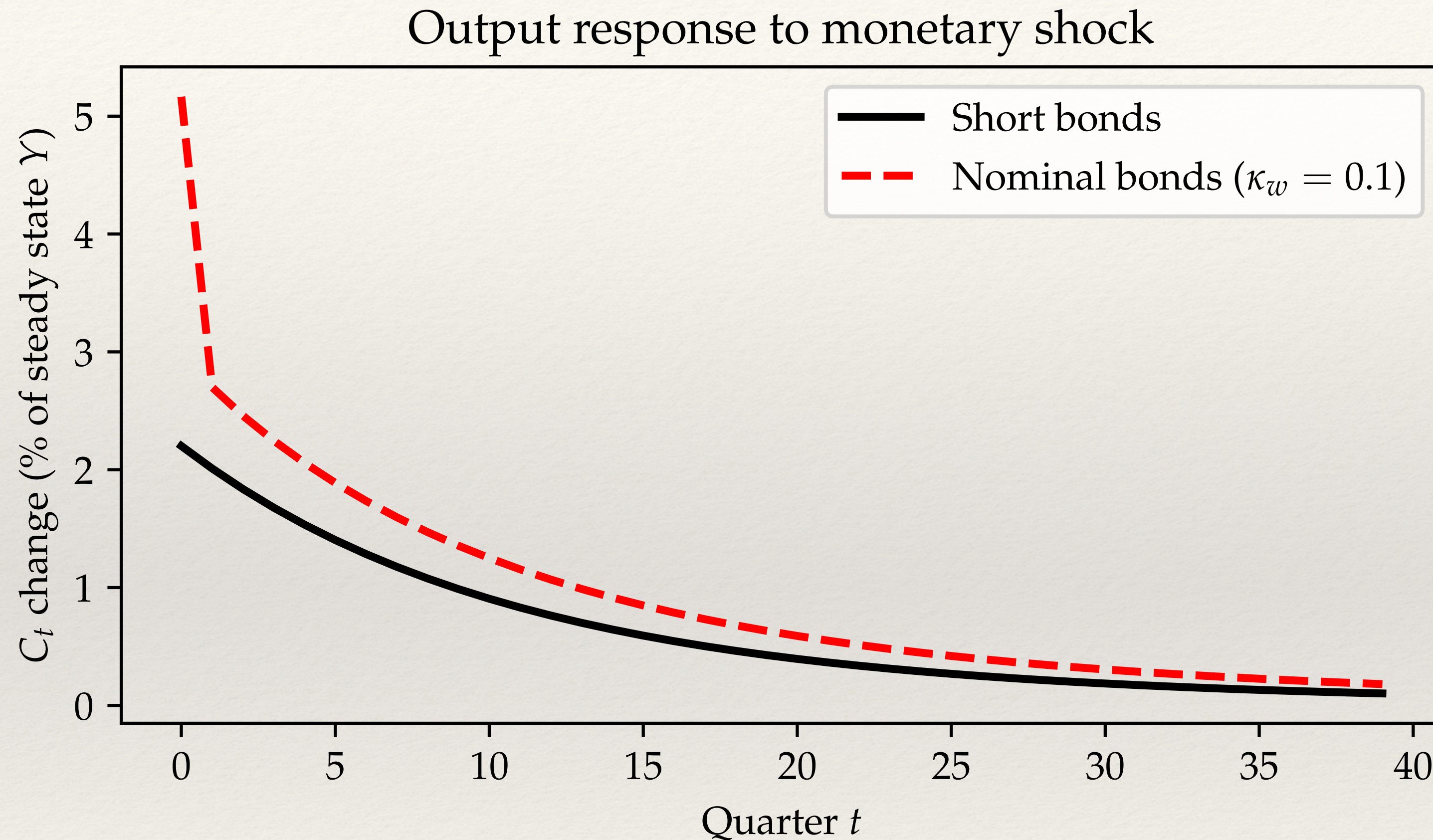
Rewriting the problem

- ❖ Now redefine $a_{it} \equiv A_{it}/P_t$. Using Fisher equation, we have

$$\begin{aligned} \max_{c_{it}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(N_t) \right) & r_{t+1}^p &= r_t, t \geq 0 \\ c_{it} + a_{it} & \leq (1 + r_t^p) a_{it-1} + Z_t e_{it} & 1 + r_0^p &= (1 + r_{ss}) \frac{P_{ss}}{P_0} = (1 + r_{ss}) \frac{1 + \pi_{ss}}{1 + \pi_0} \\ a_{it} & \geq 0 \end{aligned}$$

- ❖ Same for gov budget constraint, fiscal rule continues to keep $(1 + r_t)B_t$ constant
- ❖ Even with r_t rule, inflation now matters for aggregate demand due to nominal revaluation! Use Phillips curve: $\pi_t = \kappa_w(Y_t - 1) + \beta\pi_{t+1}$

Output effect of a monetary policy shock



- ❖ **Fisher effect:** inflation redistributes from bondholders to taxpayers
- ❖ Those agents have higher MPCs, this boosts demand (effect bigger with steeper P.C.)

The stock market

Introducing a stock market

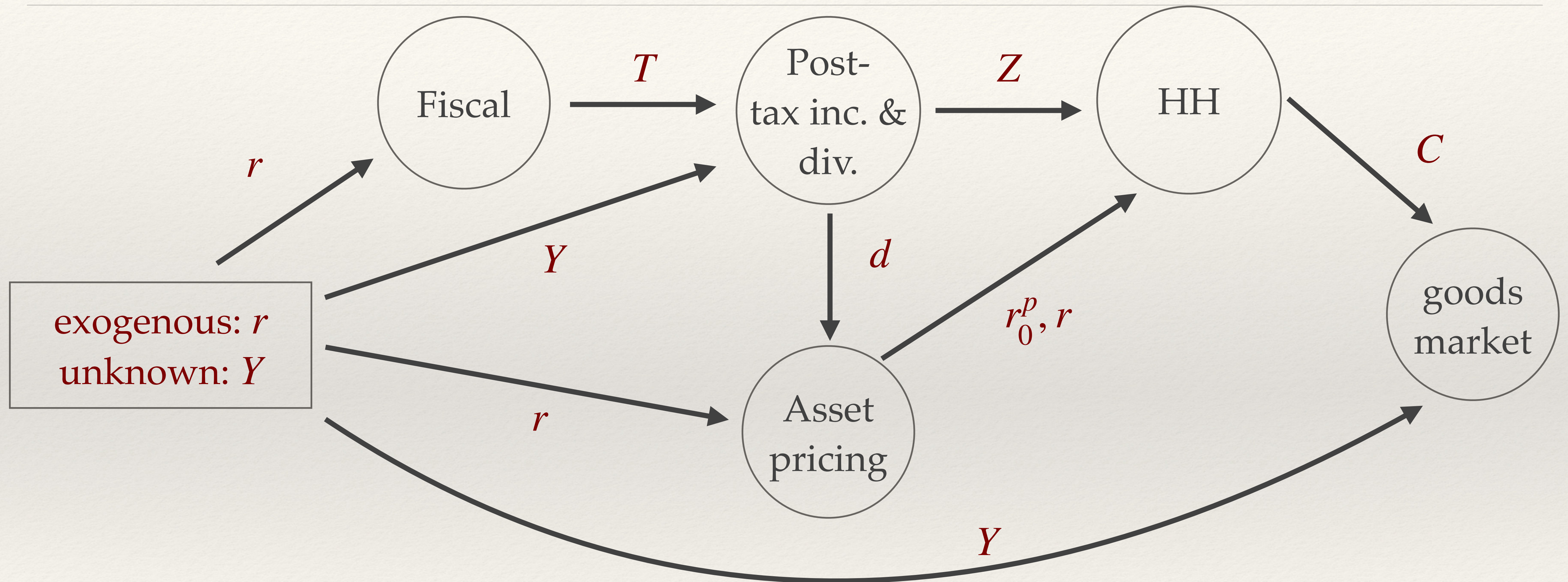
- ❖ Calibration with $A_{ss} = B_{ss} = Y_{ss}$ not realistic
 - ❖ Households hold a lot more wealth than this, ~ 5 GDP in the U.S.
 - ❖ Invested (at least) in stocks **in addition** to bonds
- ❖ To incorporate this, change calibration to have markups $\mu > 1$, $A_{ss}/Y_{ss} = 5$
- ❖ Firm problem now implies $\frac{W_t}{P_t} = \frac{1}{\mu}$ and dividends $d_t = (1 - \tau_t) \left(1 - \frac{1}{\mu}\right) Y_t$
- ❖ Stocks are claim to dividends with price p_t
- ❖ A mutual fund owns bonds and stocks, sells shares to hh's with return r_t^p

[Households owning stocks and bonds directly similar, see portfolio lecture]

Asset pricing conditions

- ❖ No arbitrage condition for the mutual fund: $1 + r_t = \frac{p_{t+1} + d_{t+1}}{p_t}$, for $t \geq 0$
- ❖ Ex-post return to mutual fund holders between t and $t + 1$: $r_{t+1}^p = r_t$
- ❖ What is the ex-post return between $t = -1$ and $t = 0$?
- ❖ Stock market price at date 0 is $p_0 = \sum_{s=1}^{\infty} \left(\prod_{u=0}^{s-1} \frac{1}{1 + r_u} \right) d_s$ in ss, just $p_{ss} = \frac{d_{ss}}{r}$
- ❖ Total ex-post return is $1 + r_0^p = (1 + r_{ss})\omega + \frac{p_0 + d_0}{p_{ss}}(1 - \omega)$
s.s. share of bonds in mutual funds portfolio B_{ss}/A_{ss}

DAG of model with traded markups

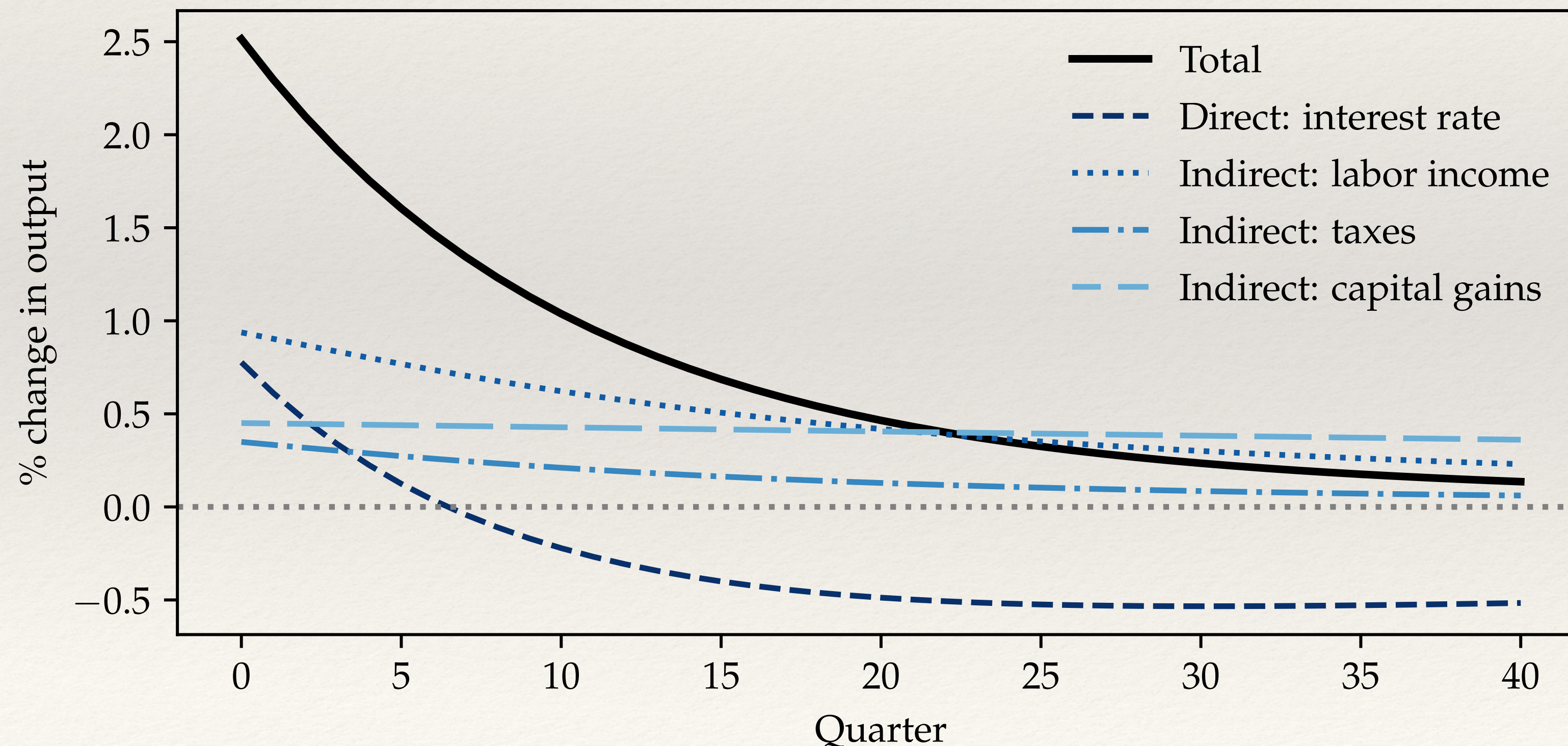


- ❖ Goods market clearing: $Y_t = \mathcal{C}_t \left(r_0^p, \{r_s\}, \{ (Y_s - T_s) / \mu \} \right) + G_{ss}$

Impulse response to AR(1) monetary shock

❖ Calibration: $B_{ss}/Y_{ss} = 1$

$$d\mathbf{Y} = \underbrace{\mathbf{M}^r dr}_{\text{Direct interest rate effect}} - \underbrace{\mathbf{M}/\mu B_{ss} dr}_{\text{Tax effect}} + \underbrace{\mathbf{M}/\mu d\mathbf{Y}}_{\text{Labor income effect}} + \underbrace{\mathbf{m} A_{ss} dr_0^p}_{\text{Cap. gain effect}}$$



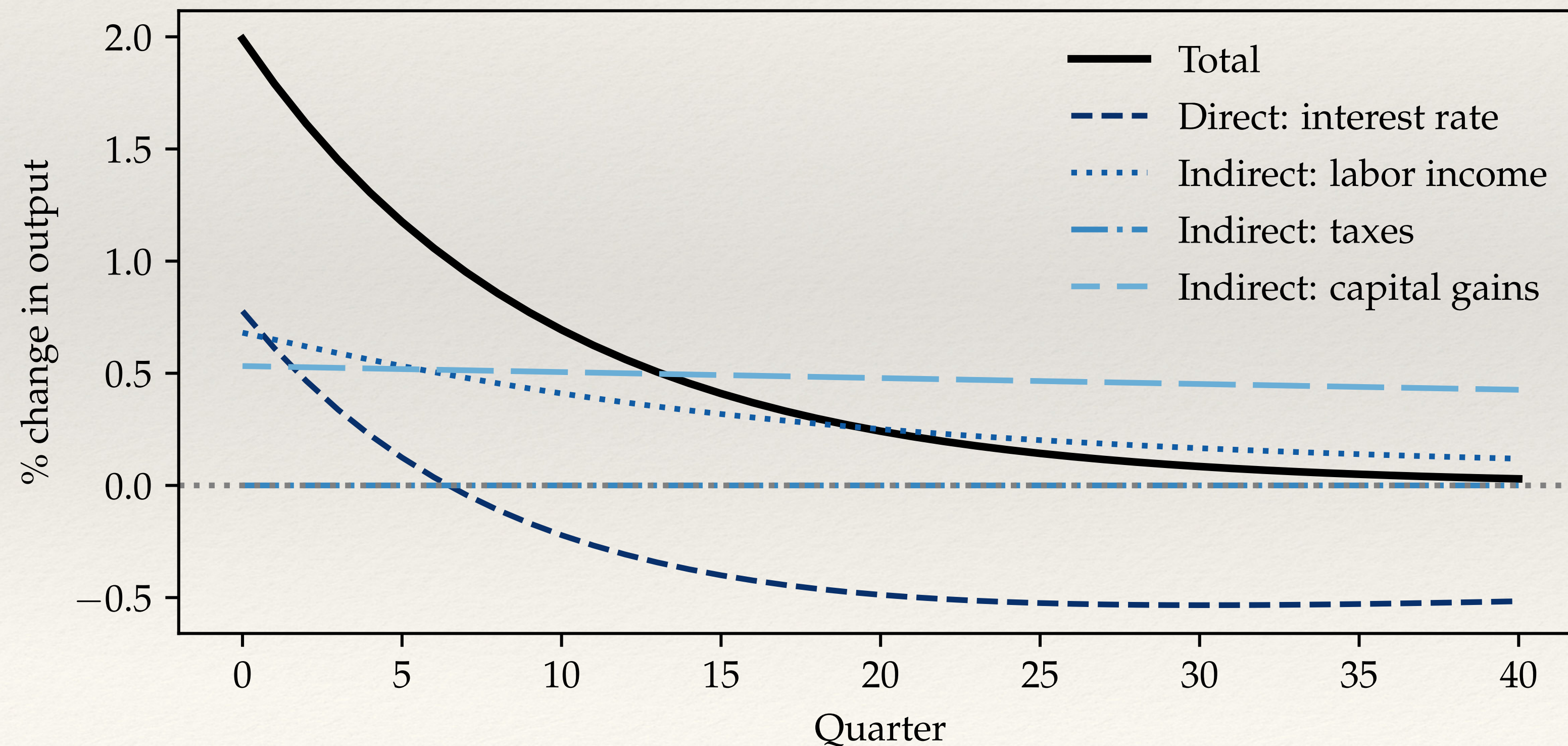
New capital gain effect!

Magnitude governed by $\mathbf{m}$,
the MPC out of capital gains
(low in data & model, here
about 1 cent on the dollar
per quarter)

Impulse response in model without bonds

❖ Calibration: $B_{ss}/Y_{ss} = 0$

$$d\mathbf{Y} = \underbrace{\mathbf{M}^r dr}_{\text{Direct interest rate effect}} + \underbrace{\mathbf{M}/\mu d\mathbf{Y}}_{\text{Labor income effect}} + \underbrace{\mathbf{m}A_{ss} dr_0^p}_{\text{Cap. gain effect}}$$



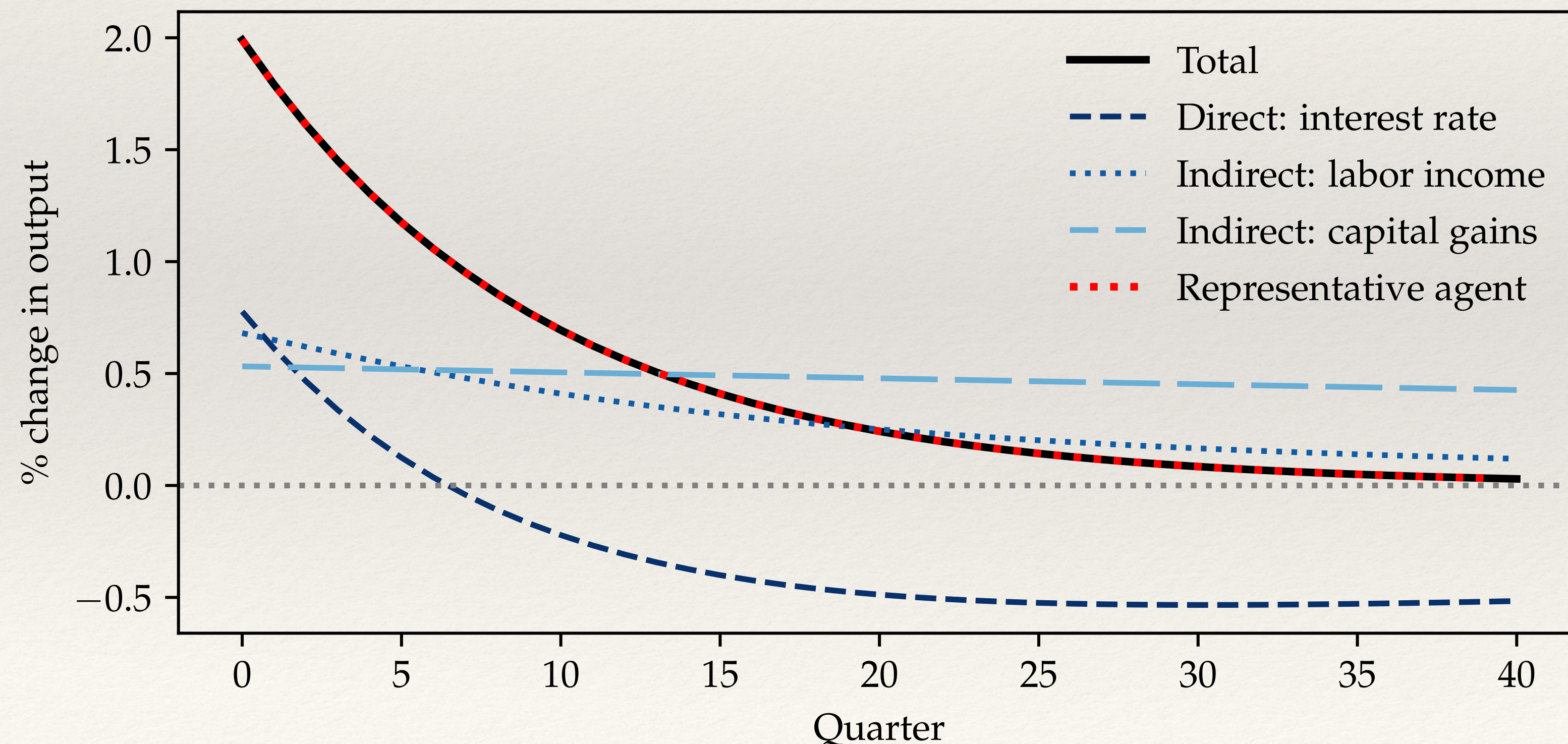
Tax effect is now zero...

Impulse response in model without bonds

❖ Calibration: $B_{ss}/Y_{ss} = 0$

$$d\mathbf{Y} = \underbrace{\mathbf{M}^r dr}_{\text{Direct interest rate effect}} + \underbrace{\mathbf{M}/\mu d\mathbf{Y}}_{\text{Labor income effect}} + \underbrace{\mathbf{m}A_{ss}dr_0^p}_{\text{Cap. gain effect}}$$

Direct interest rate effect Labor income effect Cap. gain effect



Total effect is exactly the same as in RANK!

$$dC_t = 0.25 \cdot C \cdot \frac{\rho^t}{1 - \rho}$$

Coincidence?

Neutrality result

- ❖ No! For the calibration with $B_{ss} = 0$ and $u(c) = \log c$, we prove (see ARE):

$$d\mathbf{Y} = -C\mathbf{U}d\mathbf{r} \qquad \mathbf{U} \equiv \begin{pmatrix} 1 & 1 & 1 & 1 & \dots \\ 0 & 1 & 1 & 1 & \dots \\ 0 & 0 & 1 & 1 & \dots \\ 0 & 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

- ❖ Monetary policy effects in HANK and RANK are the same for **any** $d\mathbf{r}$!
- ❖ **Werning (2015) result version 2** (log utility + “acyclical liquidity”)

Summary

- ❖ HANK substantially enriches the analysis of monetary policy
- ❖ Key points:
 - ❖ Countercyclical income risk has large amplification effects
 - ❖ Maturity structure important due to capital gains-induced redistribution
 - ❖ Nominal positions relevant due to inflation-induced redistribution
 - ❖ With traded stocks, can get other instance of equivalence to rep agent
- ❖ Topics we left out (see next lecture+ARE paper): monetary-fiscal interactions, fiscal theory of the price level, investment, Taylor rules...